1. Why LangGraph exists?
2. What is LangGraph?
3. LangChain vs LangGraph (When to use what)

Important:

1. Video will be unapologetically long
2. Prerequisite

What is LangChain

LangChain is an open-source library designed to simplify the process of building LLM based applications.

It provides modular building blocks that let you create sophisticated LLM-based workflows with ease.

LangChain consists of multiple components

1. **Model** components gives use a unified interface to interact with various LLM providers
2. **Prompts** component helps you engineer prompts
3. **Retrievers** component helps you fetch relevant documents from a vector store

But the biggest offering of LangChain is **Chains.**

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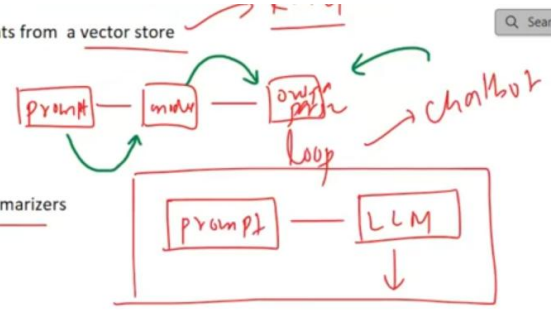
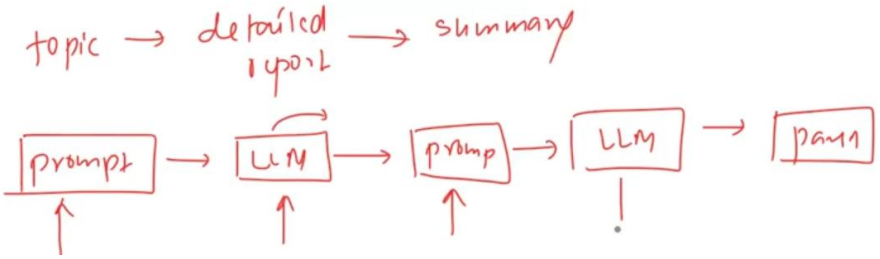


What can you do with LangChain

1. Simple conversational workflows like Chatbots, Text Summarizers
2. Multistep workflows
3. RAG applications
4. Basic level agents



Automated Hiring workflow

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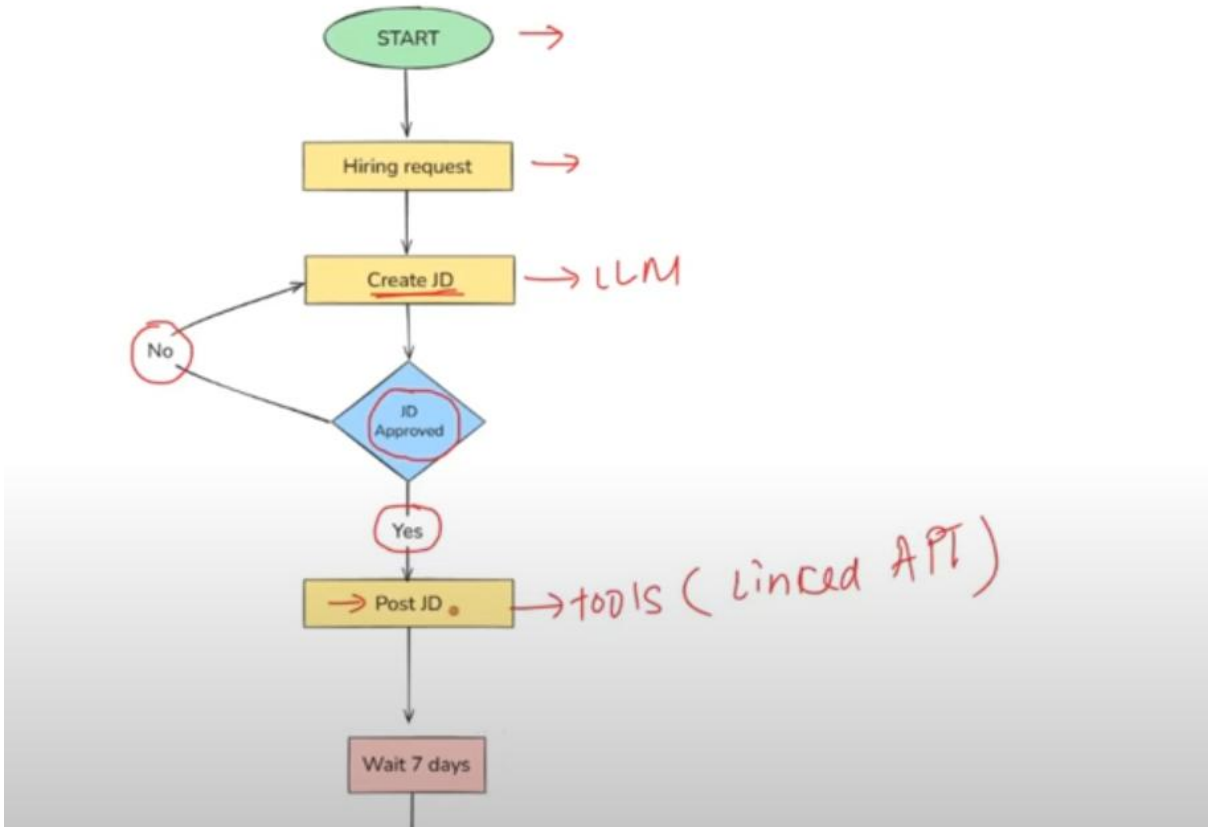

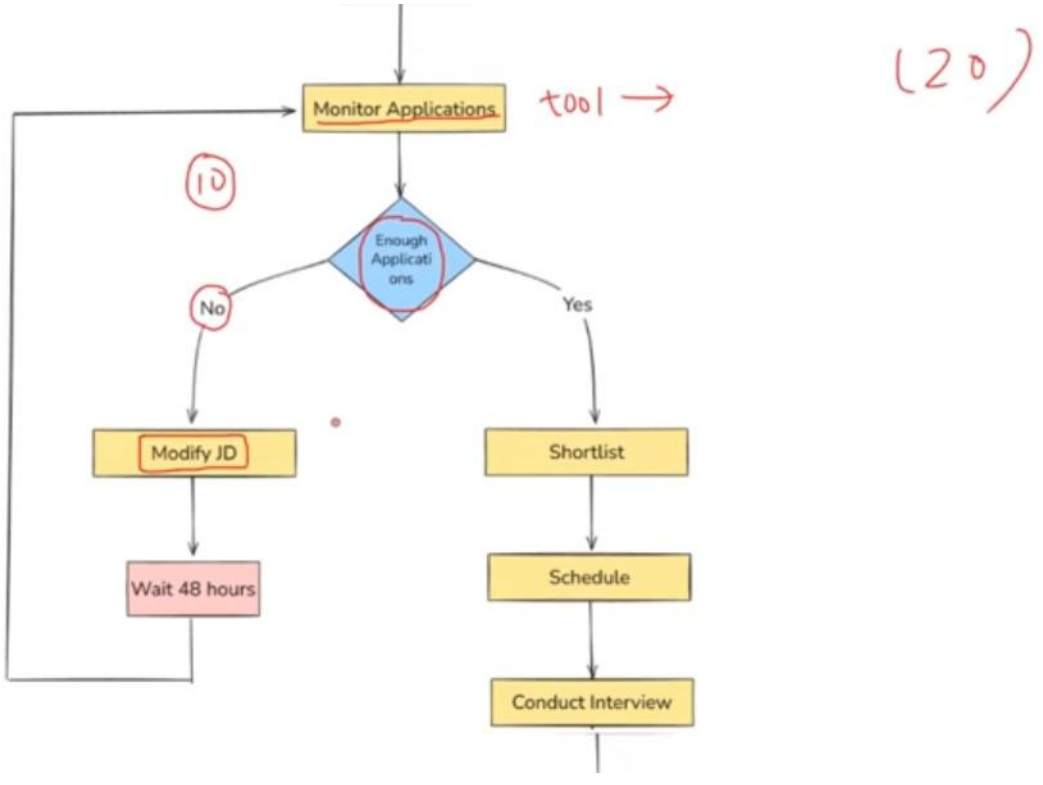

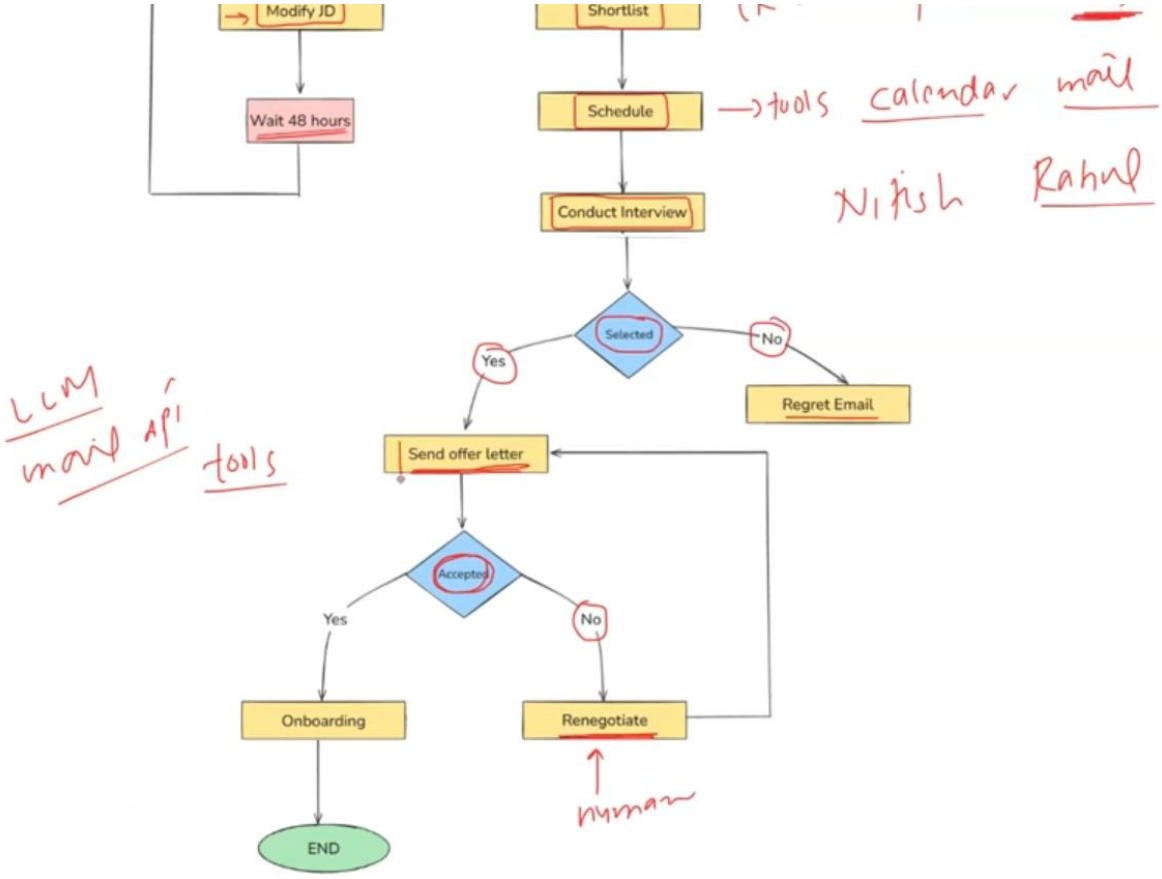

Challenges

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1. Challenge #1 - Control Flow Complexity

1) conditional branch
2) loops
3) jump

```
hiring_prompt = "We need to hire a Software Engineer for our backend team."
```

```
# Step 2: Create JD using LLM
llm = ChatOpenAI(model="gpt-4", temperature=0)

jd_prompt = ChatPromptTemplate.from_template(
    "Create a job description based on the hiring request:\n\n{request}"
)
jd_chain = jd_prompt | llm | StrOutputParser()
```

```
# Step 3: Approval function
def approve_jd(jd: str) -> bool:
    # Simulate approval logic
    return "Approved"
```

```
# Step 4: Post JD function
def post_jd(jd: str):
    print("JD Approved and Posted:\n")
```

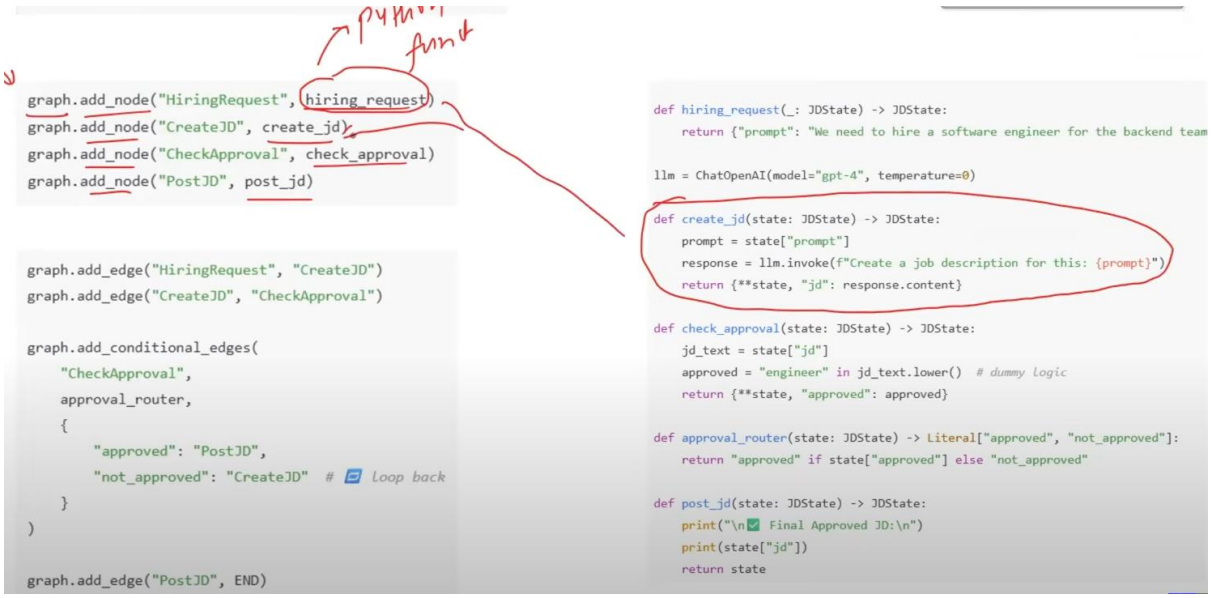
```
# Step 5: Loop until JD is approved
approved = False
jd_output = None

while not approved:
    jd_output = jd_chain.invoke({"request": hiring_prompt})
    print(jd_output)

    approved = approve_jd(jd_output)

    if not approved:
        print("JD not approved. Regenerating...\n")

# Final Step
if approved:
    post_jd(jd_output)
```



```python
graph.add_node("HiringRequest", hiring_request)
graph.add_node("CreateJD", create_jd)
graph.add_node("CheckApproval", check_approval)
graph.add_node("PostJD", post_jd)

graph.add_edge("HiringRequest", "CreateJD")
graph.add_edge("CreateJD", "CheckApproval")

graph.add_conditional_edges(
    "CheckApproval",
    approval_router,
    {
        "approved": "PostJD",
        "not_approved": "CreateJD"  # 🔁 Loop back
    }
)

graph.add_edge("PostJD", END)
```

```python
def hiring_request(_: JDState) -> JDState:
    return {"prompt": "We need to hire a software engineer for the backend team."}

llm = ChatOpenAI(model="gpt-4", temperature=0)

def create_jd(state: JDState) -> JDState:
    prompt = state["prompt"]
    response = llm.invoke(f"Create a job description for this: {prompt}")
    return {**state, "jd": response.content}

def check_approval(state: JDState) -> JDState:
    jd_text = state["jd"]
    approved = "engineer" in jd_text.lower()  # dummy logic
    return {**state, "approved": approved}

def approval_router(state: JDState) -> Literal["approved", "not_approved"]:
    return "approved" if state["approved"] else "not_approved"

def post_jd(state: JDState) -> JDState:
    print("\n✅ Final Approved JD:\n")
    print(state["jd"])
    return state
```

Challenge #2 - Handling State

```
state = {
    "goal": "Hire a backend software engineer",
    "jd": "",  # Job description text
    "jd_approved": False,
    "jd_posted": False,
    "min_applicants": 5,
    "num_applications": 0,
    "shortlisted_candidates": {
        # "candidate_id": {"name": ..., "score": ..., "status": ...}
    },
    "interview_questions": [],
    "offer_status": {
        "sent": False,
        "accepted": False,
        "renegotiated": False
    },
    "onboarding_status": {
        "completed": False,
        "start_date": None,
        "employee_id": None
    }
}
```

Challenge #3 - Event Driven Execution

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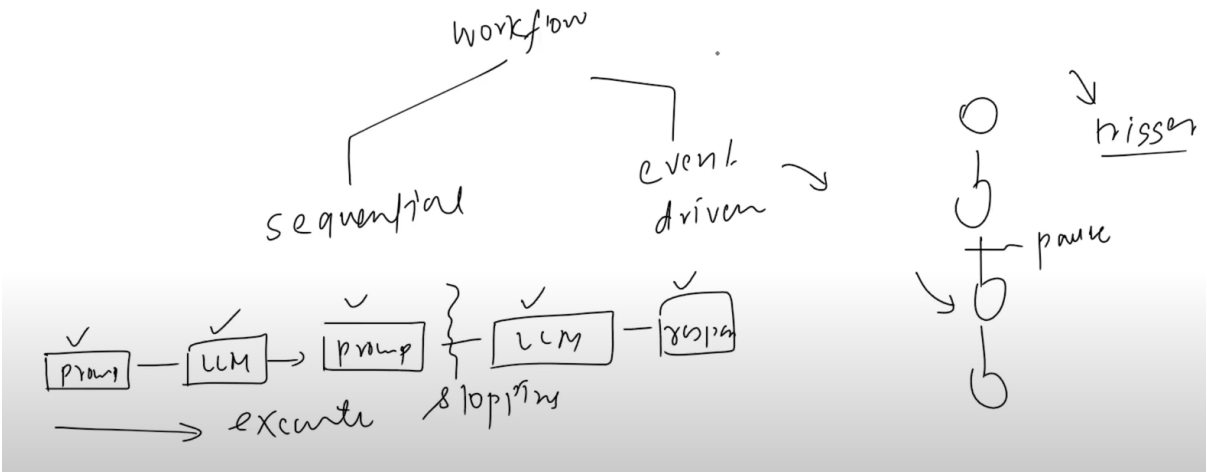
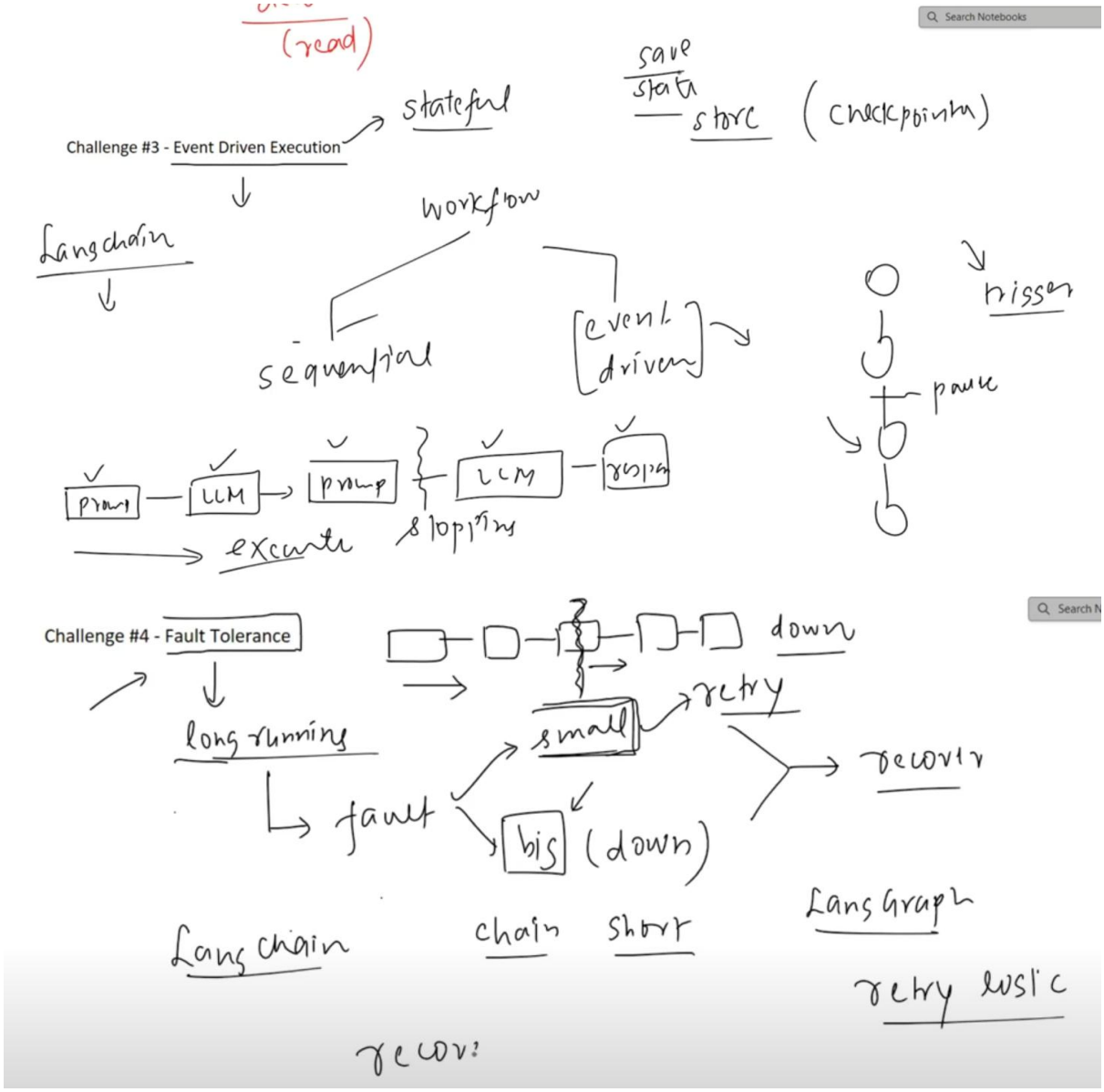


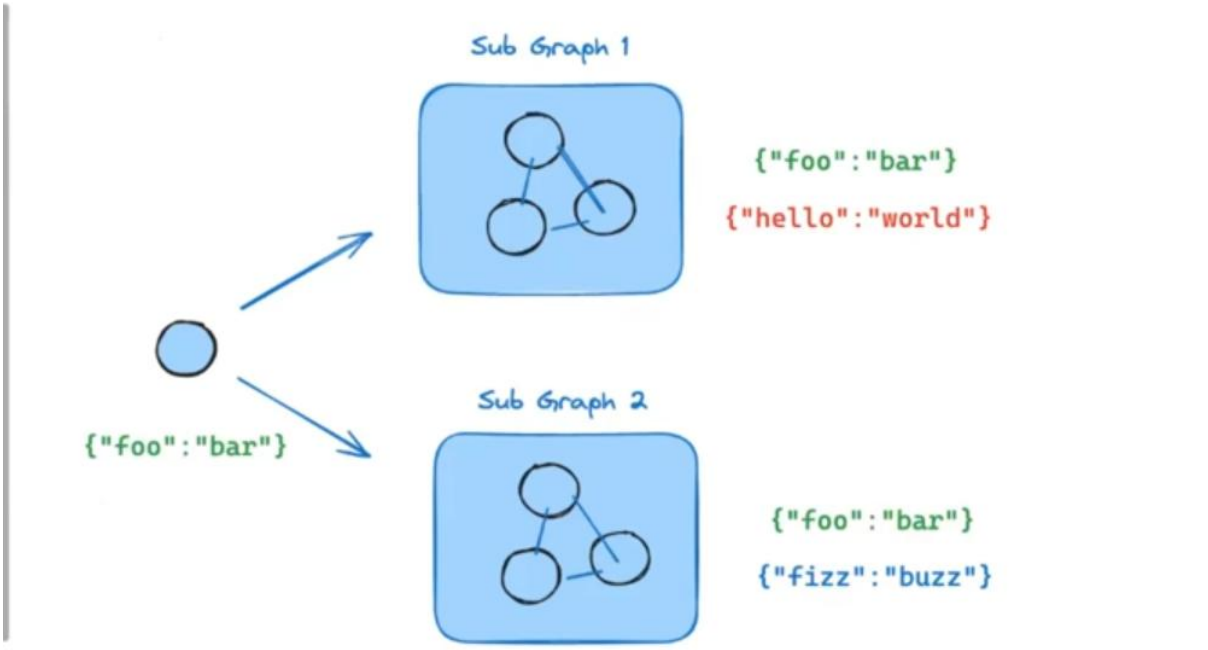


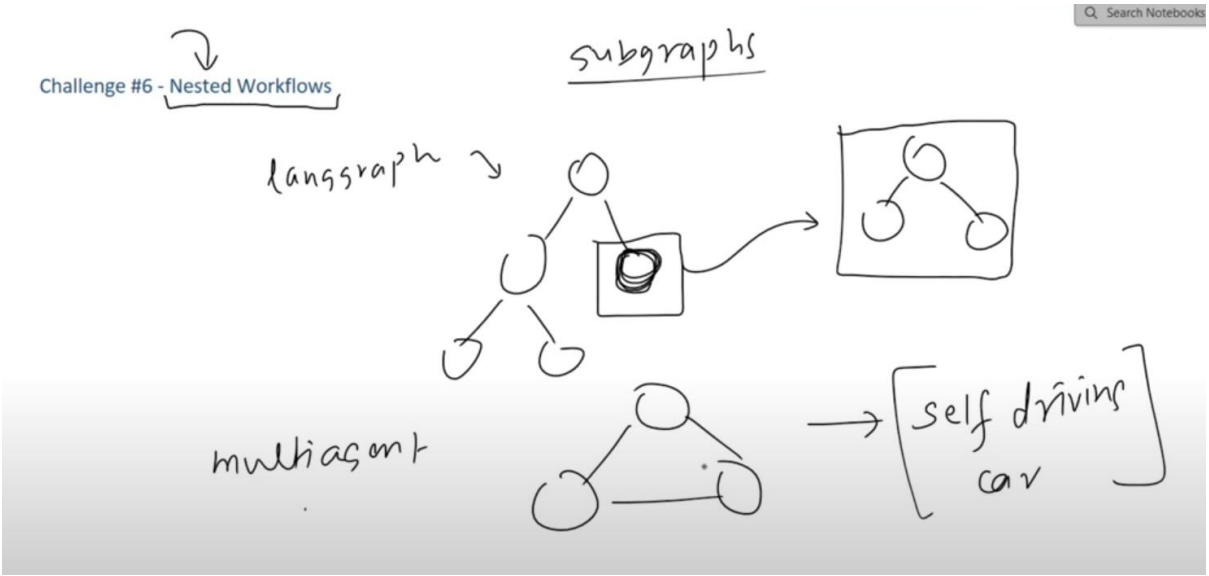
Challenge #5 - Human-in the loop

Challenge #6 - Nested Workflows

Subgraphs

A subgraph is a graph that is used as a node in another graph — this is the concept of encapsulation applied to LangGraph. Subgraphs allow you to build complex systems with multiple components that are themselves graphs.





Challenge #7 - Observability

Observability refers to how easily you can monitor, debug, and understand what your workflow is doing at runtime.

Conclusion

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1. What is LangGraph?

LangGraph is an orchestration framework that enables you to build **stateful**, **multi-step**, and **event-driven** workflows using large language models (LLMs). It's ideal for designing both **single-agent** and **multi-agent** agentic AI applications.

Think of LangGraph as a **flowchart engine for LLMs** — you define the steps (nodes), how they're connected (edges), and the logic that governs the transitions. LangGraph takes care of **state management**, **conditional branching**, **looping**, **pausing/resuming**, and **fault recovery** — features essential for building robust, production-grade AI systems.

2. When to Use What?

- Use **LangChain** when you're building **simple, linear workflows** — like a prompt chain, summarizer, or a basic retrieval system.
- Use **LangGraph** when your use case involves **complex, non-linear workflows** that need:
  - Conditional paths
  - Loops
  - Human-in-the-loop steps
  - Multi-agent coordination
  - Asynchronous or event-driven execution

3. Should We Still Use LangChain?

Yes. LangGraph is built **on top of LangChain** — it doesn't replace it.

You'll still use **LangChain components** like:

- `ChatOpenAI` (LLMs)
- `PromptTemplate`
- `Retrievers`
- `DocumentLoaders`
- `Tools`, etc.

LangGraph handles **workflow orchestration**, while LangChain provides the **building blocks** for each step in that workflow.